HDOC- 5 – 1
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Program to Convert Time in Seconds to Hours:Minutes:Seconds Format

This document provides a clean, well-commented implementation for converting a given time in seconds into the hours:minutes:seconds format in C. The program takes the total number of seconds as input and uses integer division and the modulus operator to compute the equivalent hours, minutes and seconds.

1. C Implementation

Step 1: Understand the Logic

- 1 hour = 3600 seconds and 1 minute = 60 seconds.
- Integer division (/) gives the number of complete hours or minutes.
- The modulus operator (%) gives the remaining seconds after removing hours or minutes.

Step 2: Define Variables and Data Types

- Use integer variables (int) to store the total seconds, hours, minutes and remaining seconds.

Step 3: Take User Input

- Display a prompt using printf("Enter time in seconds: "), then read the value using scanf("%d", &totalSeconds).

Step 4: Convert the Time

1. Compute hours = totalSeconds / 3600.
2. Compute minutes = (totalSeconds % 3600) / 60.
3. Compute seconds = totalSeconds % 60.

Step 5: Format the Output

- Print the result in HH:MM:SS format using printf("%02d:%02d:%02d", hours, minutes, seconds).

Complete C Code

```c
/*
 * Q5 (User Inputs, Operations & Output)
 * Write a program to input time in seconds and convert it
 * to hours:minutes:seconds format.
 */
#include <stdio.h>

int main() {
    int totalSeconds, hours, minutes, seconds;

    // Step 1: Read the total time in seconds
    printf("Enter time in seconds: ");
    scanf("%d", &totalSeconds);

    // Step 2: Convert seconds to hours, minutes and seconds
    hours   = totalSeconds / 3600;
    minutes = (totalSeconds % 3600) / 60;
    seconds = totalSeconds % 60;

    // Step 3: Display the result in HH:MM:SS format
    printf("Time in HH:MM:SS format = %02d:%02d:%02d\n",
           hours, minutes, seconds);

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc5_1.c -o hdoc5_1
$ ./hdoc5_1
Enter time in seconds: 3661
Time in HH:MM:SS format = 01:01:01
```

Run 2:

```
$ ./hdoc5_1
Enter time in seconds: 9045
Time in HH:MM:SS format = 02:30:45
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 3661
- Calculation:
  - hours = 3661 / 3600 = 1
  - minutes = (3661 % 3600) / 60 = 61 / 60 = 1
  - seconds = 3661 % 60 = 1
- Output: Time in HH:MM:SS format = 01:01:01

Sample Test Case 2:

- Input: 9045
- Calculation:
  - o hours = 9045 / 3600 = 2
  - o minutes = (9045 % 3600) / 60 = 1845 / 60 = 30
  - o seconds = 9045 % 60 = 45
- Output: Time in HH:MM:SS format = 02:30:45